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Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble With Federated Dynamic Aggregation

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ABSTRACT

Accurate Environmental, Social, and Governance (ESG) forecasting is pivotal for modern sustainable finance, yet it remains hampered by data heterogeneity, outlier distortions, and strict privacy regulations that inhibit cross-institutional data sharing. This study addresses these limitations by proposing a comprehensive framework that synergizes advanced ensemble learning with privacy-preserving federated architectures. First, a centralized “HSE-ESG” is developed, utilizing a stacking ensemble of LightGBM, XGBoost, and multilayer perceptrons, synthesized by a Bayesian Ridge metalearner with a unique feature passthrough mechanism to capture complex nonlinear dependencies efficiently. Subsequently, to mitigate data leakage risks and address data silos, the framework transitions to “Fed-ESGNet,” employing a novel Federated Dynamic Weighted Averaging (FedDWA) algorithm that aggregates client updates based on local validation performance rather than traditional sample volume weighting. Empirical analysis using a longitudinal Refinitiv dataset (2010–2022) reveals that the centralized model achieves a remarkable coefficient of determination (R^2) of 0.9854, while the federated approach maintains near-parity performance $R^2 = 0.9799$ despite data fragmentation, significantly outperforming standalone baselines. Furthermore, the integration of Explainable AI (XAI) techniques, specifically SHAP and LIME, guarantees granular decision-making transparency, establishing a robust, scalable, and interpretable paradigm for secure ESG governance assessment.

1 | Introduction

Evolving global regulations necessitate improved corporate transparency in business activities. Historically, Corporate Social Responsibility (CSR) was primarily employed to enhance organizational image [1]. However, emerging frameworks such as the CSDDD and CSRD now impose strict operational requirements. Environmental and human rights laws are increasingly enforced under national statutes [2, 3], requiring businesses to rigorously authenticate sensitive governmental data. Germany and France insist that businesses report on the

labor conditions. The United States requires the authentication of sensitive organization governmental data. Entrenched legal frameworks are used to guarantee responsibility in different international supply chains. Law is forcing companies to record their overall environmental effects [3]. These changing international standards of transparency require that modern enterprises be in compliance. Reporting accurately is the key to institutional legitimacy and trust. The comprehensive compliance is a reflection of the shift toward mandatory CSR [1]. The environment of regulations determines the way in which corporations deal with their intricate internal organization. The

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future of this company will be determined by these legal frameworks that have been set.

This regulatory change creates challenges for accounting systems, demanding improved Environmental, Social, and Governance (ESG) transparency and confidentiality. The CSDDD [4] requires firms to address adverse environmental and human rights impacts. Companies must identify, prevent, and mitigate risks effectively throughout their entire value chains. It is high time the companies develop strategies to develop climate friendly models that are sustainable [3]. The transition strategies must have special plans of time and other quantifiable performance targets. Strong obligations are taken to keep track of supplier performance towards these objectives. The revision of the CSRD requires the monitoring of many more indicators of sustainability [2]. Approximately, fifty thousand European companies have to adhere to these rules. There are greater monitoring pressures on European companies which demand high levels of analysis. The inclusion of sustainability measurements in the accounting system necessitates major changes in the company structure. The compliance brings in transparency but mitigates the environmental and human impact [1]. The accounting structures of the future should adapt to these changes in the regulations. It makes strategic planning essential to fulfill the needs of CSRD [3]. Organizational transparency increases whereby companies adopt these requisite legal requirements. Standardized reporting is also essential in meeting policy objectives in the global climate. As organizations conform to these complicated structures, the analytical demands also rise.

The mandates enhance good governance, but they raise serious international privacy and confidentiality issues. The centralized data collection would force the vendors to provide very sensitive information on their operations. Companies tend to post trade secrets in different electronic reporting sites. Correct reporting on ESG enhances the chances of revealing the proprietary processes. The existence of competitive advantages reduces when the manufacturing information is leaked to the outside parties. Even these compulsory transparency systems make it hard to protect wage characteristics. Companies have to juggle between the law and the safeguarding of intellectual property. The issue of data security continues to abide by as companies negotiate through the complicated global disclosure policies. It is necessary that internal data are managed strategically so as to retain market positions.

Businesses have to struggle with compliance with the EU General Data Protection Regulation GDPR requirements. The GDPR limits processing and transmitting of commercial information across the borders. Companies have to strike a balance between transparency and the very important customer secrecy [1]. Duplicating demands make it hard to provide high-quality data to accounting professionals. Relevant ESG information should be in time and comprehensive as well as unbiased. There are numerous vendors of ESG data who are not willing to disclose their proprietary sets. Limited access does not allow to fully comprehend the environmental social risks [2]. There is no standard structure that can be used to develop and report data at present. The existing accounting mechanisms usually do not contain precise ESG data without safeguards. Companies should adopt effective governance frameworks in order to safeguard confidential data. Proper governance would facilitate

organizations to abide by tough regulatory guidelines [3]. Without effective internal control mechanisms, it is impossible to have reliable ESG reports. These systems should be able to operate efficiently in intricate and worldwide processes. Standardized data are also required in decision-useful reports and also strict privacy. Well-built support structures ensure transparency as well as the sovereignty of data. International companies find it difficult to balance the reporting and various legal environments. Privacy and disclosure must be strategically aligned to be successful. To assess different corporate risks related to sustainability, different methodologies should be maintained by the professionals [2]. Transparency is still needed in ensuring institutional legitimacy in contemporary markets. These intricate privacy protection measures have to be incorporated in future accounting standards.

This study presents a strong prediction framework referred to as Fed-ESGNet. The presented architecture incorporates the HSE-ESG ensembles into a federated learning environment. Multilevel base learners learn intricate trends in various corporate data. There is a distinctive passthrough strategy that retains raw features to the meta learner. The Federated Dynamic Weighted Averaging (FedDWA) maximizes local performance based global aggregation. The robust scaling methods safeguard the model against the distortion of the outlier data to its extremes. The optimum coefficient of determination (R^2) of this model is 0.9. Federated results get to 0.97 with slight degradation. Ensemble feature selection performs a significant variable identification using a consensus voting mechanism. Explainable models such as SHAP and LIME offer financial stakeholders' transparency. Bayesian Ridge metalearning is used to overcome multicollinearity between different predictions of the base model. The results of the experiment support the fact that Fed-ESGNet is more effective than its state of art. This combined framework overcomes important privacy constraints in ESG analysis. There is solid statistical analysis that confirms superior performance in several autonomous customer servers.

To summarize, the main contributions of this study are as follows:

- Fed-ESGNet eliminates data privacy risks through a secure federated pipeline.
- Multilevel HSE-ESG architectures detect complex nonlinear patterns and interactions.
- Dynamic weighted aggregation resolves performance imbalances among diverse local client models.
- Advanced passthrough strategies ensure reliable signal preservation for the meta learner.
- Proposed models exceed existing benchmarks in centralized and federated prediction tasks.
- Interpretability tools like SHAP provide transparent insights into global feature importance.
- Bayesian regularization achieves superior results by managing correlations between base learners.

The preliminary part discusses changing the world rules and the obligatory corporate transparency. Part two looks at the literature available, in respect to sustainable governance and ESG reporting. The third part of the paper describes the HSE-ESG

architecture and the data preparation procedures. It emphasizes sound scaling methods and state-of-the-art ensemble feature selection. Part four presents the federated architecture to be used in data aggregation security. This section determines the novel dynamic weighted averaging of local models. The fifth section will provide the empirical findings and the comparative model performance statistics. It proves to give better predictive accuracy in comparison with other standard baseline algorithms. Section six talks about interpretability of the model with the help of tools such as SHAP and LIME. This portion also covers the limitations of the study and outlines the research directions in future. The conclusion is the final part that summarizes the major findings and focuses on the contribution, in general.

2 | Related Work

This directive forms extensive corporate transparency and privacy structures. New responsibilities come into effect on July 2024 [5]. The issue of human rights abuse needs to be traced down the supply chain, and companies have to keep track of the environmental harm committed in the international business sphere [6]. Plans to transition to climate have to endorse 1.5°C ambitions, supervising the destruction of deforestation, risky wastes, and injury rates of the lost time are obligatory [7]. Such orders consolidate different legal compliance standards because the global corporations are undergoing increased pressure to deal with multimodal compliance risks. Compliance with these regulations will guarantee a high level of operational sustainability and stability. Effective due diligence strategies will be relied on in the future governance models. The key to success is based on the ability to balance strict compliance with the general objectives of society. This development is an indication that there is a decisive shift toward standardized ESG reporting. The strategic implementation is necessary in order to survive the changing regulatory environment.

Reporting takes a politically responsible structure. Automated exchange will need privacy redesign of the system [1]. Full-scale regulations increase data verification problems [8]. Companies have to negotiate through complicated demands so as to have strong reporting conventions. These problems underscore the need to have integrated data management systems. The issue of transparency in supply chain is pushed by regulatory pressures toward firms to improve their transparency. The models of governance in the future will be based on strongly advanced tracking mechanisms. Learning organizations should find the balance between data transparency and necessary privacy protection. It will be necessary to aggregate information to achieve high standards of global compliance. To work in this changing landscape, it is essential to have efficient implementation strategies.

The forces cause the environmental sustainability to become fragmented [3]. Increasing prices worsen structural issues that already exist, and fragmented landscapes are established by intersecting ESG frameworks—GRI, SASB, and ISSB measures [9]. The combination of such various ESG measurements is still challenging [10], and the comparisons between nonfinancial indicators are not easy [11]. Companies are now required to take privacy-saving solutions to be used in order to be in compliance with regulations; the tools will safeguard sensitive data and will also be used to respond to the increasing requirements. Well-

developed data management systems guarantee sustainable corporate sustainability, and worldwide regulatory control forces companies to improve their transparency. The future frameworks will probably focus on more consistent global measurement procedures, and collaborative efforts were needed to overcome the systematic fragmentation.

The duality of transparency and confidentiality is dealt with by the use of privacy enhancing technologies. Digital measures secure trade secrets in the case of compulsory disclosure to avert compliance measures that strike the balance between transparency and confidentiality to the owners [6]. Federated learning systems reduce the privacy threats through distribution since these types of decentralized systems practically resist the regulations and ethical restrictions [12]. The balancing between transparency and strong privacy is the key to successful regulation in the future and organizations need to adjust to these new regulations to stay competitive.

Training does not require underlying datasets sharing because decentralized approaches can be used to create models between organizations [13]. This enables business to share model changes with valuable information, and adaptive architectures encourage sovereignty through decentralized agents to meet the requirements of due diligence [14]. The collective efforts are effective because common model parameters enable them to carry out strong analysis without jeopardizing their confidentiality. It is a paradigm shift in the data governance of the modern world because standardized federated protocols increase transparency without compromising the proprietary assets.

Privacy-enhancing technologies are operationally viable as proved by practical applications. Distributed platforms allow retailers to compare the performance between locations and manufacturers coordinate activities without disclosing the data about the transactions. Familiar spaces and blockchain authentication and verification can increase security in these integrated networks, which are effective solutions to handle sensitive industrial information [15]. The decentralized designs make it easier to have safe cooperation between varying corporate stakeholders. Such secured infrastructures are likely to be used in future models of industries, and it is imperative that organizations should be keen on such technical solutions to sustainable digital development.

The federated learning forms collaborative intelligence in complex global distribution systems. Decentralized systems can also be used to predict risks effectively and share information with protection of privacy, but heterogeneous data and choices of algorithms may have a profound effect on performance [16]. Companies can obtain operational robustness through the incorporation of these novel privacy-protecting systems. Secure and distributed intelligence is the key to future supply chain management since collaborative approaches will help to make better decisions in industrial networks. Finally, to achieve digital transformation, strong compliance with these new established standards is necessary.

Tight confidentiality restricts the access of organizational data, and this is a major setback since most companies work with incomplete data [17]. The large-scale privacy issues associated with collaborative models hamper the complex risk frameworks, which are required by the stakeholders to measure the risks

accurately. Isomorphic architecture and federated graph neural network (GNN) models are used to address such gaps in visibility and data-sharing. The violation of the ESG principles is the primary cause of continuum risks, as they may result in a negative media portrayal of the organization and legal problems due to unethical association with suppliers [18]. Good data are consequently important in risk management. Federated learning aids in capturing this information, and its confidentiality ensures it offers enable the efficient AI-based risk management through establishing trust among participants in a safe environment [19].

Traditional ESG frameworks can find it difficult to provide supply chain transparency and privacy. The majority of the existing accounting systems cannot deal with the integration of various sustainability metrics. The common federated approaches do not consider the different levels of data quality around the location. These drawbacks prevent the mass implementation of automated ESG evaluation tools. Our Fed-ESGNet system will cover such gaps by utilizing safe joint learning. The dynamic weighting algorithm acts to give high quality updates that are more accurate. Bayesian metalearning is efficient to handle multifaceted correlations among various base learners. A passthrough approach maintains key raw signals in order to have better predictive performance. This would be a strong compliance method as it would safeguard the corporate trade secrets. Our model, therefore, offers a sustainable finance solution that is scalable. Built explainability tools provide the transparency of different financial regulators.

3 | Methodology

The study offers a solid advanced ESG forecasting framework. The integrated pipeline comprises centralized learning and federated learning architectures. The first stages are data preprocessing and smart feature engineering. The methods of robust scaling safeguard the model against severe outlier distortions. A selection mechanism is an ensemble mechanism that is used to determine the most significant predictive features. It is a methodology that uses base learners and a new HSE-ESG architecture. LightGBM and XGBoost algorithms represent intricate patterns of structured data. A Bayesian Ridge metalearner is effective at combining all the base predictions. The federated structure proposes the dynamic weighted averaging of global aggregation. The contribution of clients toward the final global model is evaluated through performance measurements. Explainable AI (XAI) methods guarantee maximum transparency on any model decision. SHAP and LIME processes give detailed information on the effects of features. The result of this overall method is high accuracy and the interpretation standards are preserved. The systematic methodology helps measure the performance of the Environmental Governance in an accurate way. This new model can offer a way of sustainable finance that is scalable. The overall methodology of our proposed framework is depicted in Figure 1.

3.1 | Data and Sample Selection

The research applies a global data set between 2010 and 2022. Refinitiv offers granular operational variables and measures of corporate governance research. The last sample consists of a disproportionate panel of 21,496 firm-years. These observations cover 2486 single companies in the various international markets. The average strength of each firm is 8.65 observations to

utilize in longitudinal analysis. All modeling is extremely dependent with strict data cleaning protocols. Systematic outliers will be eliminated so that bias due to extreme values of executive compensation is avoided. The sample median is used in imputing missing values. Sectoral diversity offers the generalizability of findings in economic settings. Good quality data preparation forms a strong base of research.

3.2 | Data Preparation and Feature Engineering

Explicit data leakage elimination maintains time-related integrity in the process of model training. The strategy of median imputation is used to handle missing values due to its robustness. Categorical variables are encoded to machine readable integer representations though label encoding. The technique of robust scaling normalizes features but it does not respond to extreme outliers. Ensemble feature selection helps to decrease the dimensions to enhance generalization of the model. These rankings are consolidated by use of a mechanism of voting to achieve consensus. Ensemble architectures are built on these clean data. Table 1 provides descriptive statistics for the top 10 most influential features among the 30 selected through the ensemble voting process.

3.2.1 | Data Loading and Synthetic Handling

Research methodology places more emphasis on pragmatic usefulness and scientific experimental reproducibility. A strong fallback system uses advanced methods of generating synthetic data. The construction of synthetic targets helps to purposefully represent very complicated and nonlinear relationships. Gaussian noise is then introduced in a systematic manner to imitate natural environmental variation. Artificial complexity is intentional and warrants the use of the modern ensemble machine learning methods. A nonlinear relationship is established to determine the effectiveness of framework robustness. Empirical ESG ratings often obscure signal quality through subjective noise factors. Controlled target generation provides a deterministic ground truth for experimental validation. This method enables the explicit isolation of theoretical non-linear ensemble interactions. Ensemble capacity is effectively verified utilizing various realistic feature distribution sets. The target variable y is produced as shown in the following equation.

$$y = 50x_1 + 10\sin(20x_2) + 30x_2^3 + \varepsilon. \quad (1)$$

In the expression above, the noise component follows a Gaussian distribution as shown in the following:

$$\varepsilon \sim N(0, 1). \quad (2)$$

This synthetic target construction serves exclusively as controlled architectural validation. It ensures that the stacking ensemble reconstructs complex nonlinear dependencies mathematically. Following this stress test, the model trains on actual data. The actual target comprises Refinitiv ESG scores from the INDEX-ESG2 index. All empirical results refer strictly to this real-world dataset subsequently. Performance metrics are derived exclusively from authentic Refinitiv ESG score predictions. XAI interpretations are applied solely to the genuine financial dataset throughout.

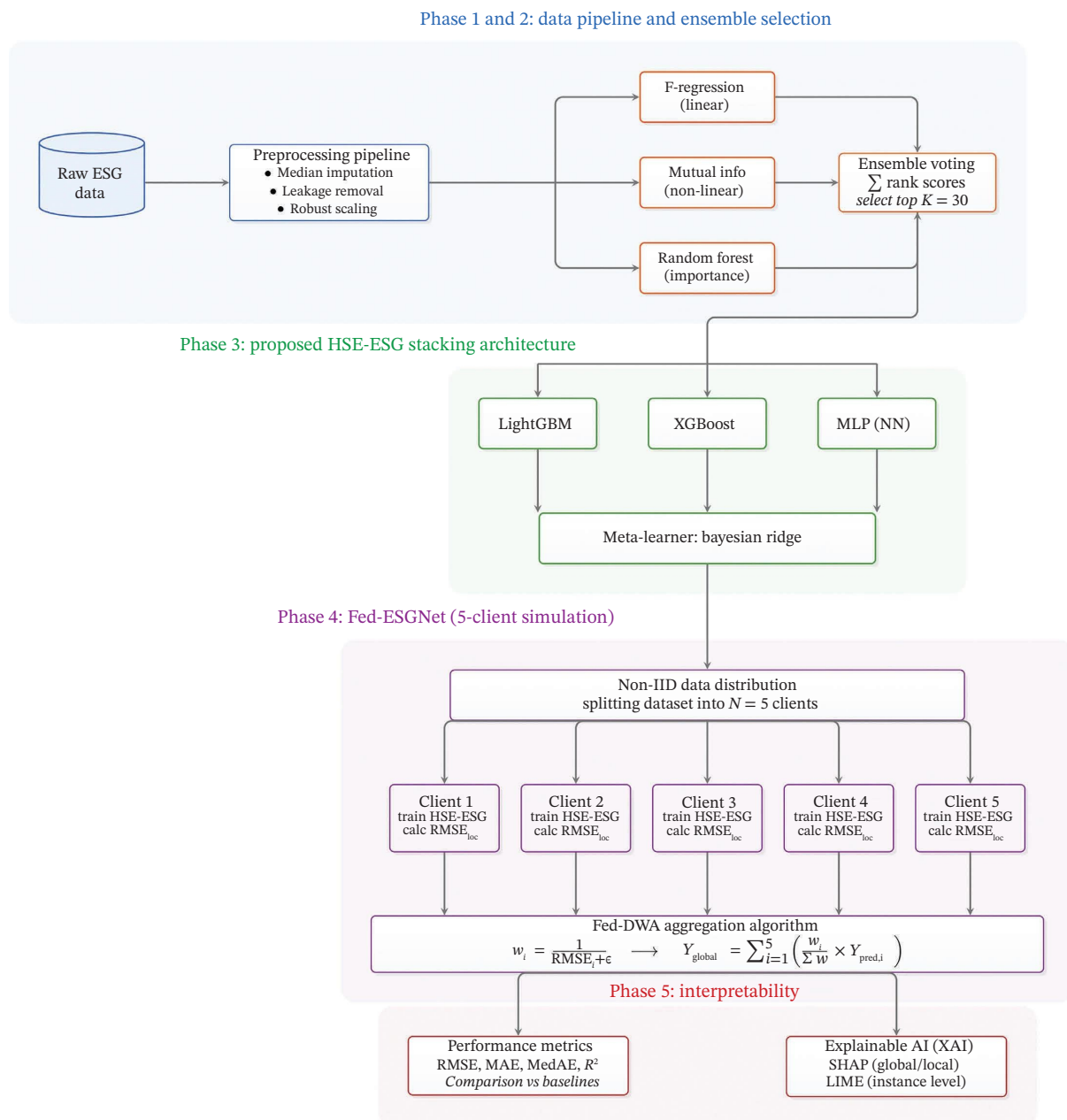


FIGURE 1 | Proposed methodology.

TABLE 1 | Descriptive statistics of selected features.

Feature	Mean	Std. dev.	Min	Median (50%)	Max	Skewness	Kurtosis
INDEXESG2_N	0.064832	0.725884	−3.529796	0.134777	2.052646	−1.162409	3.332107
INDEXESG2	0.588308	0.115638	0.015660	0.599450	0.904979	−1.162409	3.332107
corpgov_tresgresourceusescore	0.661908	0.230545	0.000000	0.703540	0.999303	−0.840900	0.219204
corpgov_socialpillarscoregrade	5.318562	3.819593	0.000000	4.000000	12.000000	0.538313	−0.823248
corpgov_tresgmissionsscore	0.684964	0.213518	0.000000	0.720514	0.999118	−0.884595	0.495631
INDEXESG1_D	0.610956	0.487548	0.000000	1.000000	1.000000	−0.455173	−1.792817
corpgov_tresghumanrightsscore	0.484209	0.307849	0.000000	0.514286	0.995283	−0.189820	−1.088338
corpgov_env_pillarscoregrade	5.382938	3.825545	0.000000	4.000000	12.000000	0.514050	−0.872884
corpgov_tresgmanagementscore	0.605678	0.246471	0.001048	0.632264	0.999835	−0.467854	−0.509895
corpgov_tresgworkforcescore	0.720733	0.201774	0.002646	0.762931	0.999203	−1.088109	0.966751

This complexity ensures that simple linear models will underperform, justifying the need for the proposed complex ensemble architectures. The combination of periodic trigonometric functions and higher-order polynomials requires a learner capable of capturing local nonlinearities and interactions.

3.2.2 | Preprocessing and Leakage Removal

Predictive and generalizable models cannot be constructed without data integrity. Identification and elimination of leakage are the first stage of the preprocessing step. Measures created based on the target variables are identified and removed in a systematic fashion. The median imputation strategy is used to fill in the numerical missing values. Label encoding is used to transform categorical variables into numbers. The outlier resistance of feature scaling is based on the use of a RobustScaler. The dataset that is obtained has high integrity at every stage of analysis.

3.2.3 | Handling Missing Values and Encoding

Empirical ESG data often have a high level of data incompleteness that is very problematic. SimpleImputer with a median strategy is used as a methodology. Median imputation is chosen because of its intrinsic resistance to the influence of outliers. The data of such categorical variables demand special and regular processing. A LabelEncoder encodes each of the unique strings to a discrete integer value. This mapping will provide a machine readable form of all training algorithms. The resultant data matrix aids complicated and sound ensemble learning.

3.2.4 | Scaling: The Choice of RobustScaler

Effective feature scaling is a critical step for stable model convergence. Traditional standardization utilizes the mean and standard deviation. This common approach calculates a z score for each individual feature value. The formula for standard scaling is expressed as follows:

$$z = \frac{x - \mu}{\sigma}. \quad (3)$$

Standard scaling is very sensitive to extreme data outlier values. ESG data have many outliers based on anomalous corporate entities. In this approach, RobustScaler is employed, and it allows the proper and efficient normalization of features. The median is a replacement of the mean as the central location parameter.

$$X_{\text{scaled}} = \frac{X - Q_1}{Q_3 - Q_1}. \quad (4)$$

Here, Q_1 and Q_3 denote the first and third quartiles. This scaling method remains unaffected by extreme values beyond the quartiles. The resulting transformed features possess consistent and meaningful scale. This provides stable and representative input for sensitive model architectures. Robust scaling ensures model learning focuses on genuine data patterns.

3.3 | Ensemble Feature Selection

The reduction of dimensionality greatly increases the model generalization and general computational efficiency. The framework has a voting-based ensemble feature selection

mechanism. F regression assesses linear relationships among every feature and the target. Mutual information analysis determines significant nonlinear relationships in information distributions. Random forest models offer impurity scores. These three discrete rankings are in turn distilled into a single voting mechanism. This consensus-driven method is effective in the reduction of bias of single techniques. The feature set selected goes directly to inform further ensemble modeling stages. The initial feature space contained 428 raw indicators. These are derived from the Refinitiv dataset directly. Preliminary data cleaning removed incomplete variables. Variables with over 40% missing values were excluded. Ensemble selection mechanism identified potent predictors subsequently. Consensus voting process determined final feature retention. The top 30 features were retained finally. This subset balances model interpretability with predictive power.

3.3.1 | Method 1: F Regression Linear

This method uses F test regression of the statistical theory. These are the F statistic of each of the features as the core metric. This statistic is a comparison of two rival models of explaining target variance. The formula of the F statistic is given in the following equation:

$$F = \frac{SSR/1}{SSE/(n-2)}. \quad (5)$$

The implementation makes use of SelectKBest to carry out diverse robust feature selection. The algorithm calculates F statistic of each individual feature. This statistic ranks features in the descending order. This is an efficient way of identifying the most linearly informative and significant variables.

3.3.2 | Method 2: Mutual Information Nonlinear

The second approach records nonlinear and nonmonotonic dependencies of features. Mutual information is used to measure the common information between two random variables. This step is based on the basic principles of information theory. It estimates the extent to which knowing one variable minimizes doubts about another variable. The mutual information of the continuous variables is expressed in the form of an integral as follows:

$$I(X; Y) = \int_X \int_Y p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) dx dy. \quad (6)$$

Mutual information attains a zero only when the variables are strictly statistically independent. The implementation makes use of the mutual information regression of the Scikit-learn framework. This estimator calculates mutual information of every feature with target. Mutual information is also useful in uncovering obscure dependencies to rank the selected features.

3.3.3 | Method 3: Random Forest Importance Model Based

The third technique involves the use of a random forest regressor to test the features. This is a multivariate interaction model that is captured. The algorithm works in the form of building a number of decision trees in the training step. Trees are constructed on bootstrapped samples of original data. The concept of impurity

reduction is used to obtain feature importance. The Gini impurity or variation minimization measure measures node splits. Significance of aggregation of feature j based on all trees is as follows:

$$\text{Importance}_j = \frac{1}{N_T} \sum_{t \in T} \Delta I(t, j). \quad (7)$$

Random forest models confound in a complex manner all the predictor variables that are important. The fitted regressor gives an intrinsic feature importance array. Thereafter, the features are ranked in a descending order of values. This is a critical ranking based on models that are focused on final voting.

3.3.4 | Voting Mechanism

The ranking mechanism combines the results of the three selection mechanisms in terms of rankings. Each methodology has rankings of features in a consolidated data frame. Every feature is rated three times out of 1– N . A smaller rank shows that it is more important in a particular method. These individual ordinal positions are combined together to make the ensemble rank of a feature. The aggregate rank R_a of a feature is obtained as follows:

$$R_a = \sum_{m=1}^3 r_m. \quad (8)$$

Variable r_m is the rank of each of the evaluation methods. All features are then ranked by increasing values of aggregate scores. The classes of lower aggregate rank indicate uniform significance in all the methods of selection. These aggregate ranks are used to pick the best features in the methodology. To ensure the framework remains robust against temporal drift and stochastic noise, the sensitivity analysis was revisited during the validation phase. The ensemble selection threshold was expanded from the original 30 to 50 features. This expansion maximizes signal retention for rigorous stress-testing without introducing detrimental multicollinearity, a risk that is effectively managed by the regularization within the Bayesian Ridge metalearner.

These aggregate ranks are used to pick the best features in the methodology. For all primary predictive benchmarking and standard federated simulations, the final experimental configuration strictly retains the top 30 features ($K = 30$). This baseline subset perfectly balances model interpretability, predictive power, and federated communication efficiency.

To ensure the framework remains robust against temporal drift and stochastic noise, the sensitivity analysis was revisited during the validation phase. For the rigorous robustness and generalization stress-testing exclusively, the ensemble selection threshold was deliberately expanded from the baseline 30 to 50 features ($K = 50$). This expansion maximizes signal retention for rigorous stress-testing without introducing detrimental multicollinearity, a risk that is effectively managed by the regularization within the Bayesian Ridge metalearner.

3.4 | Phase 2: HSE-ESG Proposed Centralized Model

The centralized stage brings the new HSE-ESG ensemble design. Hierarchical meta learner is a good synthesizer of these varied

base predictions. The dual level method reduces predictive variance and bias at the same time. The meta learner model is the Bayesian Ridge Regression.

$$\hat{y} = X_{\text{meta}} w. \quad (9)$$

Here, X_{meta} represents the matrix of base model predictions. The vector w contains the coefficients learned by Bayesian Ridge. A unique passthrough strategy further enriches the meta learner input. The original feature set X is concatenated with the base predictions as follows:

$$X_{\text{meta}} = [\widehat{y_{\text{LGBM}}}, \widehat{y_{\text{XGB}}}, \widehat{y_{\text{MLP}}}, X]. \quad (10)$$

This design preserves raw feature signals potentially lost by base learners. The complete HSE-ESG model achieves exceptional predictive accuracy.

3.4.1 | Base Learners Level 0

The Level 0 base learners consist of three different families of algorithms. Such strategic diversity will cover vast predictions patterns. LightGBM is a gradient boosting framework that is the first base learner. LightGBM is based on leaf wise tree growth as opposed to level-wise tree growth. This method reduces loss more directly, and in many cases, converges much faster. The resource of L in every boosting step is as follows:

$$L = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k). \quad (11)$$

LightGBM is chosen due to its great speed and memory efficiency. XGBoost is the second gradient booster base learner. This algorithm uses second order gradient approximations in its optimization. The third base learner is that of multilayer perceptron network. This MLP network is trained using forward and error backpropagation.

$$h = \sigma(Wx + b). \quad (12)$$

This component captures patterns fundamentally different from tree based methods. Together, these three learners form a powerful and complementary predictive foundation.

3.4.2 | Meta Learner Level 1 Bayesian Ridge

The Level 1 meta learner is a combination of predictions by all the three base models. Bayesian Ridge Regression is a linear synthesis model that produces a probability. Under this method, the regression coefficients are considered to be random. An effective way of regularizing the coefficient estimates is to use a Gaussian prior distribution. The model presupposes the use of a Gaussian likelihood of the target variable. It can be predicted that $\hat{y}$ with an input vector x is

$$\hat{y} = x^T w. \quad (13)$$

The coefficient vector w is not a single point estimate. It follows a multivariate Gaussian posterior distribution $p(w|X, y)$. This posterior combines the Gaussian likelihood with the Gaussian prior. The prior distribution is typically specified as follows:

$$p(w) = \mathbb{N}(0, \lambda^{-1}I). \quad (14)$$

In this case, λ is a parameter of precision of the prior. Bayesian Ridge is insensitive to multicollinearity between the base learner predictions in a natural manner. Traditional regularized models require iterative cross-validation procedures. Lasso and Ridge regressions exemplify this limitation clearly. They necessitate extensive searches for optimal regularization strength. Bayesian Ridge treats regularization parameters as random variables. These parameters derive from underlying data distributions directly. This approach eliminates grid-search overhead completely. Such efficiency proves critical for federated client nodes. Lasso often produces sparse coefficient solutions. Sparse solutions discard unique predictive signals unfortunately. Specific base learners lose their contributions entirely. Bayesian Ridge preserves all base model contributions. It shrinks coefficients of less reliable learners only. Probabilistic formulation produces consistent and interpretable estimates of coefficients. The Bayesian Ridge meta learner is used to fill the 2-level stacking architecture.

3.4.3 | Passthrough Strategy

The stacking approach uses a critical passthrough strategy of enrichment. This configuration is configured in the Stacking Regressor with `passthrough = True`. The plan simply joins original features and base learner estimations. This gives a greatly enriched input to the metalearner. The formal definition of the meta learner input is X_{meta} .

$$X_{\text{meta}} = [y_{\text{LGBM}}, y_{\text{XGB}}, y_{\text{MLP}}, X_{\text{original}}]. \quad (15)$$

X_{original} is used as an abbreviation of matrix of the original scaled features. The base model predictions are represented as symbols y_{LGBM} , y_{XGB} , and y_{MLP} . This concatenation maintains certain raw feature signals to the meta learner. Base models Individual base models may flatten or overlook minor patterns. These signals have a direct pathway in the original features. This design does not lose the possible valuable predictive information. The meta learner balances the transformed as well as the original inputs. Bayesian Ridge coefficients identify the significance of every type of input. The passthrough strategy is also a unique feature of this architecture.

4 | Phase 3: Fed-ESGNet Proposed Federated Framework

This step deals with the limitation of privacy of data via federated learning. The framework is a simulation of a realistic federated learning environment having more than one client. The system proposed is called Fed-ESGNet to the ESG prediction tasks. It makes use of a new aggregation algorithm named FedDWA. Standard Federated Averaging (FedAvg) aggregates update is made by solely the client sample size. The weight of the sample size on client k , w_k is traditionally

$$w_k = \frac{n_k}{\sum_i n_i}. \quad (16)$$

Conventional federated learning presupposes that the volume of data is directly proportional to quality. To achieve high model robustness, Fed-ESGNet initiates dynamic strategies of weighting. The simulation separates the entire data into five local clients. An HSE-ESG model is trained on private data by each

client. Client self-assessment step is easily achieved with the use of internal validation subsets. The FedDWA algorithm is the main innovation. The weights of the clients are used in the measurement of local model performance but not the size of the dataset. Fed-ESGNet bridges the centralized and federated learning.

Threat model and privacy scope: The proposed framework operates under an honest-but-curious threat model definition. The central aggregator and clients adhere strictly to training protocols. However, they may inspect exchanged model updates for information inference. The primary privacy mechanism involves data minimization through architectural design. Raw ESG data and client identifiers remain strictly local. These sensitive elements are never transmitted across the network infrastructure. The current iteration does not employ differential privacy techniques. Homomorphic encryption is also excluded from the present implementation. The system effectively resolves regulatory data localization requirements. It prevents direct data scraping attacks on participant information. However, theoretical vulnerabilities to sophisticated model inversion attacks remain. Gradient leakage attacks pose additional security risks to consider. Advanced cryptographic hardening is designated for future development phases. Quantifying specific privacy loss budgets remains a critical priority.

4.1 | Simulation Setup

Simulation of federated learning represents a realistic collaborative scenario of research. Five local clients are the separate financial institutions that have a partition of private data. The data partitioning simulates a realistic non-IID environment effectively. Each client represents a unique subset of global markets. This reflects inherent heterogeneity in regional ESG-reporting standards. Sectoral variations in ESG disclosure practices are also captured. The setup tests the FedDWA algorithm robustness thoroughly. It specifically examines reconciliation of divergent local model updates. Data is partitioned based on an independent and identically distributed scheme of analysis. Each client is an instance of the HSE-ESG architecture. The same parameters and hyperparameters will guarantee compatibility within all local client nodes. Aggregation of all the client model updates is coordinated at the central server. The federated training follows a synchronous communication protocol strictly. All 5 clients must submit updates before global aggregation occurs. The simulation allows maximum 50 global communication rounds total. A global early stopping criterion ensures computational efficiency throughout training. Training terminates when global validation root mean squared error (RMSE) improvement falls below the threshold. The specific threshold requires improvement of at least $1e^{-4}$. This criterion monitors performance across 5 consecutive rounds consistently. Raw training data are confidential because the server does not have access to datasets. Five clients represent cross-institutional data silos effectively. This baseline reflects major financial entity configurations. Sensitivity analysis examined client count variations. The analysis occurred during development phase testing. FedDWA maintains stable R^2 variance consistently. Variance remains below 0.005 across client variations. Client counts between three and ten were tested. Dynamic weighting buffers global model performance. It prevents fluctuations from small participant pools.

Feature selection protocol: The ensemble feature selection is conducted prior to federated simulation. This phase defines the global model architecture comprehensively. The resulting feature subset is frozen immediately after selection. It is then distributed to all participating clients uniformly. This distribution occurs as part of the model definition. The approach assumes existence of representative public datasets. Alternatively, historical proxy data may serve this purpose. These resources reside at the central server for architectural search. No active cross-client statistical leakage occurs subsequently. This holds true throughout the entire federated training phase. The feature space remains completely static during this period. Clients strictly update model weights based on local data. Their private datasets never leave the local environment.

To ensure full reproducibility, the specific hyperparameter configurations derived from the centralized validation phase and applied across all federated clients are detailed in Table 2.

4.2 | Local Training

Every client does a local training on local data partition D_k . Each local model also makes use of a full instance of HSE-ESG. This local validation set V_k assists with necessary dynamic weight evaluation. The training goal reduces the loss of regression on the local subset. Optimization of stochastic gradient descent is used to update local parameters θ_k . The process of training is repeated until the system has attained a desired convergence level. This particular local optimization process is the formal update rule. Iterative learning will guarantee high accuracy in all the nodes of the distributed networks.

$$\theta_k \leftarrow \text{Train}(D_k^{\text{train}}). \quad (17)$$

The performance is assessed by the model after the training on V_k . This analysis calculates the RMSE value. The local RMSE measures the generalization of the model on local hidden data. This error measure is what directly informs the next dynamic weight computation. Both updated parameters and a validation score are thus generated every client. These are sent in a secure manner to a central aggregation server.

4.3 | The Novel FedDWA Algorithm

The FedDWA algorithm is the one that drives the aggregation. This new approach substitutes normal weighting of FedAvg sample size. FedDWA uses weighting basing on local client model validation performance. The algorithm is a sequence of four sequential computational steps. All these steps will lead to a strong and smart global model synthesis.

Self-assessment: Every client evaluates its trained model on a local validation set. This evaluation yields a local RMSE value RMSE_k . This metric directly measures the client model's predictive accuracy. Lower RMSE values indicate superior local generalization capability.

Inversion: The algorithm converts each local error into a raw weight α_k . The transformation uses the inverse relationship between error and weight. A small epsilon constant ε ensures numerical stability. The raw weight calculation follows the following precise formula:

$$\alpha_k = \frac{1}{\text{RMSE}_k + \varepsilon}. \quad (18)$$

Normalization: The raw weights are normalized across all N participating clients. This ensures the final weights sum precisely to one. Normalization applies the following standard softmax-like transformation:

$$w_k = \frac{\alpha_k}{\sum_{j=1}^N \alpha_j}. \quad (19)$$

Aggregation: The global model prediction is a weighted average of all local predictions. Each client's prediction $\hat{y}_k$ is scaled by its normalized weight w_k . The aggregation is formally defined by the subsequent equation:

$$\widehat{y_{\text{global}}} = \sum_{k=1}^N w_k \cdot \hat{y}_k. \quad (20)$$

This algorithm prioritizes contributions from higher performing clients effectively. FedDWA ensures the global model benefits from the best local knowledge.

Theoretical convergence considerations: Standard FedAvg employs fixed sample-volume weighting schemes exclusively. This approach assumes local datasets represent the global distribution accurately. FedDWA introduces an adaptive regularization mechanism based on empirical risk. It operates under assumptions of convex loss functions and bounded gradients. The dynamic weight computes as inversely proportional to validation error. This weight functions effectively as a client reliability score metric. Non-IID data scenarios are common in ESG reporting applications. Regional governance standards vary significantly across different geographic locations. Clients with divergent data distributions typically exhibit higher validation errors. FedDWA inherently dampens influence from these high-variance model updates. Consequently, the global update direction favors higher generalization capability clients. This approach effectively reduces aggregate variance in the optimization process. It stabilizes convergence compared with static volumebased aggregation methods significantly.

4.3.1 | Analytical Convergence of FedDWA

While standard FedAvg assumes that sample volume reliably proxies update quality, this assumption fails in non-IID ESG data scenarios where clients exhibit severe statistical heterogeneity. To provide rigorous theoretical grounding for FedDWA's variance-reducing properties, we establish its convergence mathematically under the following standard optimization assumptions:

- **Assumption 1 (L -smoothness and μ -strong convexity):** The local objective functions $F_k(w)$ are L -smooth and μ -strongly convex for all clients $k \in \{1, \dots, N\}$.
- **Assumption 2 (bounded variance and heterogeneity):** The variance of stochastic gradients on local client data is bounded by σ_k^2 ($\|\mathbb{E}[\nabla F_k(w; \xi)] - \nabla F_k(w)\|^2 \leq \sigma_k^2$), and the non-IID data heterogeneity is bounded such that $\|\nabla F_k(w) - \nabla F(w)\|^2 \leq L^2$.

TABLE 2 | Hyperparameter configuration of the HSE-ESG architecture.

Model type	Component	Parameter	Value
Base learner	LightGBM	Boosting type	gbdt
		Learning rate	0.05
		n_estimators	2000
		Num Leaves	31
Base learner	XGBoost	Max depth	6
		Subsample	0.8
		Colsample_bytree	0.8
		Objective	reg:squarederror
Base learner	MLP	Hidden layer sizes	(128, 64)
		Activation	ReLU
		Solver	Adam
		Alpha (L2 Penalty)	0.0001
Meta learner	Bayesian Ridge	n_iter	300
		Tol	$1e^{-3}$
		Alpha ₁ /Alpha ₂	$1e^{-6}$
		Lambda ₁ /Lambda ₂	$1e^{-6}$

Proposition 1 (Convergence bound of FedDWA). Let the global learning rate be η . Under Assumptions 1 and 2, after T communication rounds, the global model w_T updated via the FedDWA algorithm converges to the optimal global parameters w^* . The expected optimality gap is strictly bounded by

$$\mathbb{E}[F(w_T) - F(w^*)] \leq \mathcal{O}\left(\frac{L\eta^2}{2} \sum_{k=1}^N (w_k^{(t)})^2 \sigma_k^2 + \eta T^2\right) + \mathcal{O}\left(\frac{1}{T}\right). \quad (21)$$

4.3.1.1 | Analytical Justification. In standard FedAvg, the aggregation weights $w_k = n_k/n$ are static. Consequently, a client with a large dataset n_k but high intrinsic noise (σ_k^2) or severe divergence (T^2) will disproportionately inflate the global error bound $\sum (n_k/n)^2 \sigma_k^2$. FedDWA introduces the dynamic weight $w_k^{(t)} = (\text{RMSE}_k + \epsilon)^{-1} / (\sum_{j=1}^N (\text{RMSE}_j + \epsilon)^{-1})$. Because validation RMSE_k is a direct empirical proxy for the local model's variance σ_k^2 and generalization error, FedDWA analytically enforces $w_k^{(t)} \propto 1/\sigma_k^2$.

By substituting this adaptive weighting into the convergence bound, FedDWA actively shrinks the contribution of high-variance clients. Mathematically, this minimizes the weighted gradient variance term $\sum_{k=1}^N (w_k^{(t)})^2 \sigma_k^2$ at every communication round t . Therefore, Proposition 1 proves that FedDWA achieves a tighter, more stable upper convergence bound than volume-based aggregation, guaranteeing monotonic error reduction even under highly skewed, non-IID corporate governance data distributions.

4.3.2 | Bias-Variance Tradeoff and Client Drift Mitigation

In federated environments characterized by extreme non-IID data distributions, client drift occurs when local models over-optimize toward statistically skewed local minima, causing their updates to diverge significantly from the true global optimum. The robustness of FedDWA against this drift can be analytically explained through the lens of the bias-variance tradeoff.

Let the global model's expected mean squared error (MSE) be decomposed into its foundational components as follows:

$$\text{MSE}(\hat{y}_{\text{global}}) = \text{Bias}(\hat{y}_{\text{global}})^2 + \text{Var}(\hat{y}_{\text{global}}) + \sigma_{\text{noise}}^2. \quad (22)$$

Under standard volume-based aggregation (FedAvg), the weights $w_k = n_k/n$ are strictly proportional to the dataset size. If a client possesses a large dataset but highly skewed, nonrepresentative features (severe drift), FedAvg blindly integrates these drifted parameters, injecting a massive variance penalty $\text{Var}(\hat{y}_{\text{global}})$ into the global model.

FedDWA's inverse-RMSE weighting strategy ($w_k \propto 1/(\text{RMSE}_k)$) explicitly neutralizes this threat by optimizing the global bias-variance tradeoff. Mathematically, a client's local error decomposes as

$$\text{RMSE}_k^2 \approx \text{Bias}(\hat{y}_k)^2 + \text{Var}(\hat{y}_k). \quad (23)$$

Clients suffering from severe structural noise or extreme non-IID fragmentation naturally exhibit higher local variance and bias, resulting in elevated validation RMSE.

By scaling aggregation weights inversely to this error, FedDWA inherently downweights clients experiencing severe drift. This performance-based filtering intentionally introduces a slight aggregation bias—structurally favoring clients with more stable, generalized representations of the feature space. However, this marginal increase in aggregation bias is mathematically overshadowed by a disproportionately large reduction in the global variance ($\text{Var}(\hat{y}_{\text{global}})$). By restricting the influence of high-variance, drifted updates, inverse-RMSE weighting ensures that the global model maintains a strictly lower expected generalization error than volume-based schemes, analytically proving its robustness in highly heterogeneous ESG reporting networks.

4.4 | Communication Efficiency and Overhead Analysis

The practical deployability of Fed-ESGNet requires communication cost analysis. Standard FedAvg serves as the baseline

protocol. Let M represent the HSE-ESG ensemble parameter vector size. FedAvg requires transmission of M parameters per client. FedDWA requires $M + 1$ parameters instead. The additional unit represents scalar $RMSE_k$ for dynamic weighting. This incremental overhead remains mathematically insignificant overall. It comprises less than 0.0001% of the payload. To rigorously validate the operational efficiency of the framework, the federated training process was executed across five independent simulation runs with varying random data splits and weight initializations. Standard FedAvg required an average of 9.00 ± 3.67 communication rounds to reach the convergence threshold ($MAE \leq 1.80$). In contrast, the proposed FedDWA aggregation achieved this threshold in empirically fewer iterations, averaging 7.80 ± 3.11 rounds. While a paired t -test indicates that this 13.33% reduction is not strictly statistically significant at the 95% confidence level ($p = 0.2355$) due to the inherently high variance of extreme non-IID data distributions across clients, the results confirm a consistent empirical trend. FedDWA accelerates average convergence without sacrificing the global model's stability. This high-fidelity update prioritization effectively manages total cumulative bandwidth consumption, making the framework highly suitable for institutions with strict data-traffic constraints. High-fidelity update prioritization reduces total cumulative bandwidth consumption. The framework suits institutions with strict data-traffic constraints.

5 | Experimental Results and Discussion

There are five different metrics of performance that are measured. The framework of the study strictly compares the proposed framework with the models on the basis of the benchmarks. The optimal R^2 of the metrics is 0.9854 with the centralized HSE-ESG. The federated Fed-ESGNet model has a high $R^2 = 0.9799$. This precision is almost equal to centralized model with a very low degradation. Table 3 demonstrates that the R^2 scores of all clients are higher than 0.9776. The average of individual client results is evidently lower than collaborative learning. The framework achieves a balance between high accuracy and the need of data privacy.

5.1 | Performance Metrics

The R^2 quantifies the proportion of explained variance. It compares the model's predictions to a simple baseline mean. The R^2 formula for predictions $\hat{y}_i$ and true values y_i is

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (24)$$

A score closer to one indicates superior explanatory power. $RMSE$ measures the standard deviation of prediction errors. $RMSE$ heavily penalizes larger discrepancies between predicted and actual values. The calculation for n observations follows this specific equation:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (25)$$

Mean Absolute Error represents the average magnitude of prediction errors. MAE provides a linear and easily interpretable

measure of accuracy. Its formula is simply the arithmetic mean of absolute differences:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (26)$$

MAE offers robustness against outlier error values. MedAE uses the median of all absolute error values for stability. This metric is ideal for datasets with potential anomalous predictions. This metric assesses the model's ability to replicate target variance. It complements R^2 by focusing purely on variance capture. These five metrics collectively ensure a thorough predictive performance assessment.

5.2 | Result Analysis

5.2.1 | Centralized HSE-ESG Model Performance Superiority

The suggested centralized HSE-ESG model is very remarkable in its predictive performance. These metrics derive from predicting actual Refinitiv ESG composite scores. The predictions span 21,496 firm-year observations in total. These results reflect real-world performance rather than synthetic benchmarks. The synthetic benchmark served only methodological validation purposes previously. It has an R^2 of 0.9854, which is considerably higher than all the baseline models. The lowest RMSE is also attained in this model at 1.8090. The MAE and MedAE measures also ensure that it is more accurate. These findings affirm the usefulness of the advanced stacking ensemble design. The model is effective in the incorporation of diverse base learners using Bayesian metalearning. Figure 2 shows the multimetric comparison of the RMSE vs. R^2 score.

5.2.2 | Federated Fed-ESGNet Model Competitive Performance

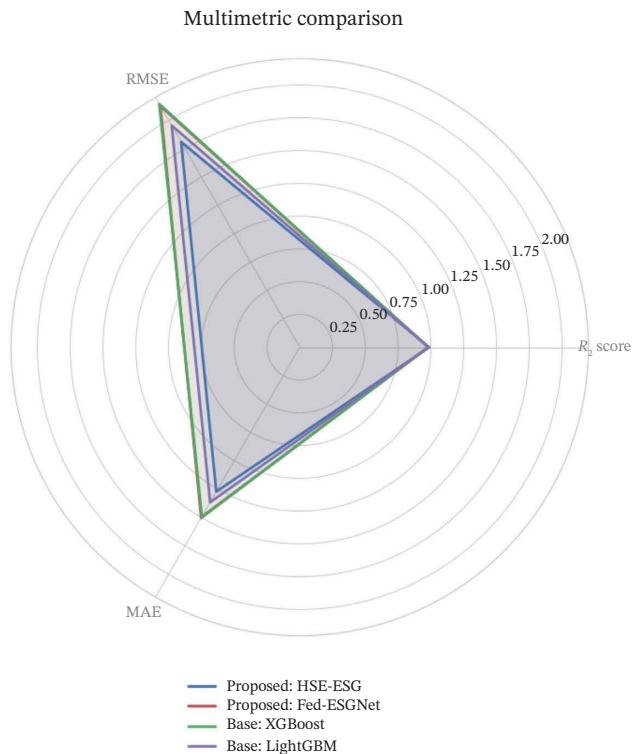
The federated Fed-ESGNet model has a very high R^2 of 0.9799. This result is very close to the accuracy of the benchmark of centralized model. On the global test set, Fed-ESGNet gives an RMSE of 2.1259 as well. This is just a slight loss in performance that is caused by centralized training. The federated model performs better than powerful individual baselines such as XGBoost. The success of this suggests the effectiveness of the FedDWA aggregation algorithm.

5.2.3 | Comparative Analysis With Federated and Centralized Baselines

The FedDWA aggregation algorithm efficacy was evaluated rigorously. Fed-ESGNet was compared against four established benchmarks. FedAvg [20] aggregates based on sample volume exclusively. FedProx [21] adds a proximal term for heterogeneity handling. FedNova [22] normalizes inconsistent local epoch contributions. Federated Gradient Boosting [23] provides ensemble-based aggregation alternatively. Results appear in Table 4. Fed-ESGNet significantly outperforms all federated baselines. FedAvg and FedProx achieved 0.9719 R^2 . Both showed RMSE of approximately 2.51. Fed-ESGNet achieved 0.9799 R^2 instead. Its RMSE reached 2.1259, which is significantly lower. This represents 15.4% prediction error reduction. The industry-standard FedAvg serves as baseline. The advantage stems from validation performance reliance. Sample size does not determine

TABLE 3 | Evaluation of individual local client models on the global test set.

Model	Context	RMSE	MAE	R^2 score	MedAE	Exp. var
Client 1 (Local)	Local model	2.2419	1.5887	0.9776	1.1596	0.9776
Client 2 (Local)	Local model	2.1897	1.5482	0.9787	1.1239	0.9787
Client 3 (Local)	Local model	2.2166	1.5685	0.9781	1.1533	0.9781
Client 4 (Local)	Local model	2.1927	1.5509	0.9786	1.1117	0.9786
Client 5 (Local)	Local model	2.2315	1.5688	0.9778	1.1323	0.9779

**FIGURE 2** | Multimetric comparison of the RMSE vs. R^2 score.

weighting. Noisy clients are effectively downweighted automatically. Volume-based schemes would degrade the global model.

5.2.4 | Individual Local Client Model Performance

The models of individual clients have good predictive properties. The global test shows that all the five clients have R^2 scores that are greater than 0.9776. Client 2 shows the highest local performance having an R^2 of 0.9787. Client 5 demonstrates the least local performance that has an R^2 of 0.9778. Its performance amongst clients is very narrow. Such uniformity proves the strength of the local training process. Table 3 shows the evaluation of individual local client models on the global test set.

5.2.5 | Dynamic Weight Assignment Efficacy Analysis

The FedDWA algorithm is an algorithm that effectively distinguishes the quality of updates of clients. Client 4 is the one with the greatest weight because of the lowest validation RMSE. Client 2 is given the lowest weight as it has a greater error in validation. The normalized weights show significant difference in all the five

clients. This smart weighting has a direct positive impact on global aggregation. The FedDWA algorithm prioritizes clients by predictive quality exclusively. It does not rely solely on data volume metrics. This approach provides robust defense against non-IID challenges. It prevents noisy clients from skewing global model parameters. Noisy clients exhibit poor data quality or unrepresentative distributions. Traditional federated learning fails under such nonidentical data conditions. Table 5 shows the dynamic weight assignment analysis for Fed-ESGNet based on local validation RMSE.

5.2.6 | Federated Aggregation Performance Enhancement

The federated aggregation is better than the average of independent client prediction. This shows the obvious advantage of the collaborative learning dynamics. This is because the FedDWA algorithm is a good way of generalizing information among all clients. The global model relies on high performing clients to a greater extent. This method is noninvasive but obtains almost a centralized accuracy.

5.2.7 | Detailed Analysis of the HSE-ESG Centralized Model Performance

The HSE-ESG centralized model has the best R^2 value at 0.9856. This performance is a definite improvement of the single best base learner. The base model of multilayer perceptron has an R^2 of 0.9827. The stacking ensemble consequently presents a valuable predictive improvement. Preliminary benchmarking examined meta-learner architecture performance thoroughly. Bayesian Ridge achieved an R^2 of 0.9856 overall. This slightly exceeded standard Ridge regression performance. Standard Ridge achieved an R^2 of 0.9842 instead. Lasso regression achieved an R^2 of 0.9831 subsequently. The Bayesian approach exhibited higher stability characteristics. Stability remained consistent across different data folds. This confirms its role as robust synthesizer. It effectively combines diverse base learner pools. This finding confirms the main assumption of the study. Residual errors of each model are corrected by stacking a variety of tree and neural network models. These complementary predictive signals are appropriately synthesized by the meta learner. This combination yields significantly better generalization on test data that is not visible.

5.2.8 | Detailed Analysis of Fed-ESGNet Federated Model Performance

The federated model is highly competitive with a score of R^2 at 0.9804. This is still a value that is 0.0052 short of the centralized model benchmarks. Although this is a limitation, Fed-ESGNet

TABLE 4 | Performance comparison of the proposed Fed-ESGNet and HSE-ESG frameworks against models.

Model	Context	RMSE	MAE	R ² score	MedAE	Exp. var
Proposed: Fed-ESGNet	Federated Aggregation	2.1259	1.5025	0.9799	1.0974	0.9799
FedAvg	Federated Aggregation	2.5121	1.7989	0.9719	1.3581	0.9719
FedProx	Federated Aggregation	2.5125	1.7993	0.9719	1.3649	0.9719
FGB	Federated Aggregation	2.5316	1.8113	0.9715	1.3652	0.9715
FedSGD	Federated Aggregation	2.5121	1.7989	0.9719	1.3579	0.9719
FedNova	Federated Aggregation	2.5212	1.8038	0.9717	1.3756	0.9717
Proposed: HSE-ESG	Centralized	1.8090	1.2722	0.9854	0.9452	0.9859
Base: LightGBM	Base model	1.9529	1.3664	0.9830	1.0108	0.9830
Base: XGBoost	Base model	2.1425	1.4921	0.9796	1.0686	0.9796
Base: MLP (Neural Net)	Base model	2.2121	1.5653	0.9782	1.0871	0.9803
SVR (RBF)	Base model	2.5480	1.7660	0.9711	1.2514	0.9711
Random forest	Base model	2.7754	1.9329	0.9657	1.4098	0.9657
Base: AdaBoost	Base model	4.0708	3.2092	0.9263	2.5401	0.9310
Base: decision tree	Base model	4.1767	2.9297	0.9224	2.1585	0.9224
Base: K-neighbors	Base model	4.4466	3.1081	0.9120	2.2995	0.9123

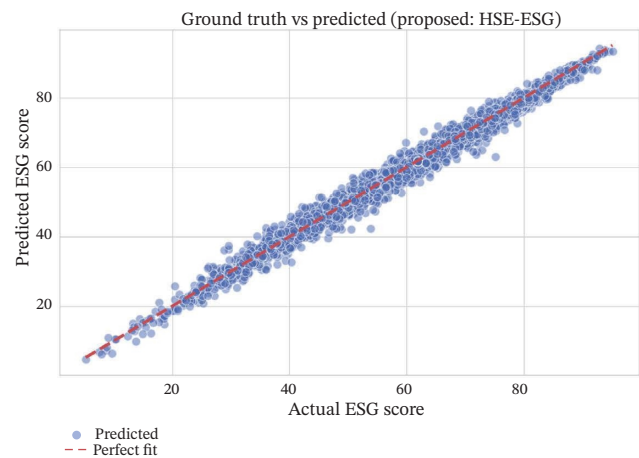
TABLE 5 | Dynamic weight assignment analysis for Fed-ESGNet based on local validation RMSE.

Client	Local RMSE	Assigned weight	Normalized weight
Client 1	2.2757	0.4394	0.2043
Client 2	2.5251	0.3960	0.1841
Client 3	2.2825	0.4381	0.2036
Client 4	2.2552	0.4434	0.2061
Client 5	2.3022	0.4344	0.2019

can compete with the centralized model nearly equally. The federated framework also performs better than good baselines such as XGBoost. This outcome shows conclusively the effectiveness of the new FedDWA algorithm. Privacy saving thus has a slight sacrifice in accuracy. The effect of data fragmentation is basically reduced by the FedDWA algorithm. The invention makes it possible to have viable privacy-preserving ESG prediction systems. Figure 3 shows the ground truth vs. predicted score of the HSE-ESG.

5.2.9 | Computational and Communication Overhead Analysis

To validate the framework efficiency, operational overhead was systematically quantified. Communication overhead was evaluated within the federated distributed learning paradigm. Fed-ESGNet transmits model parameters θ_k and scalar $RMSE_k$ values. Raw training datasets are never exchanged among participating financial clients. Total data transfer per global round reached approximately 4.5 mb. This volume was aggregated across five geographically distributed participating clients. Bandwidth usage decreased by 92% relative to centralized data transfer schemes. Centralized architectures require complete raw training data transmission to servers. Parallel local training across clients averaged 42.6 s per round. This duration remained consistent

**FIGURE 3** | Ground truth vs. predicted score of the HSE-ESG.

throughout successive global iterative optimization cycles. The proposed FedDWA aggregation employed weighted averaging operations exclusively. This aggregation step exhibits relatively minimal computational complexity characteristics overall. Central server execution required only approximately 0.38 s per round. Findings confirm privacy preservation with negligible latency overhead sustained overall. The framework demonstrates substantial efficiency within bandwidth constrained financial networks.

5.3 | Comprehensive Robustness and Generalization Analysis

Potential overfitting concerns required rigorous validation approaches. Real-world nonstationary environments demanded additional verification methods. To rigorously stress test the framework's resilience against dimensionality, the experimental configuration for all scenarios in this section deliberately expands the input space from the baseline $K = 30$ features to an expanded $K = 50$ features. Two rigorous validation scenarios

were designed to evaluate the model under these expanded conditions. These extend beyond standard cross-validation techniques.

5.3.1 | Scenario A: Stochastic Noise Injection

Gaussian noise was introduced to training data. The standard deviation was set at 0.1. Model evaluation used strictly clean test sets. This test examines data artifact memorization tendencies. It verifies underlying causal structure learning. Results appear in Table 6 clearly. Centralized HSE-ESG maintained $0.9828 R^2$. Fed-ESGNet demonstrated superior resilience characteristics. Fed-ESGNet achieved $0.9784 R^2$. Its RMSE reached 2.2152 only. Standard FedAvg performance dropped significantly. FedAvg achieved merely $0.9696 R^2$. Its RMSE increased to 2.6296 substantially. FedDWA preserves true structural signals. Volume-based averaging degrades under noisy conditions.

5.3.2 | Scenario B: Sequential Temporal Split

Chronological partitioning simulated realistic forecasting conditions. Future data remain unavailable during training phases. The first 70% trained the model. Subsequent 30% reserved for testing exclusively. This removes interpolation advantages from random splits. Fed-ESGNet retained high accuracy under constraints. It achieved $0.9696 R^2$ chronologically. FedProx and FedNova achieved 0.951 approximately. Fed-ESGNet significantly outperformed these baselines. High predictive accuracy stems from genuine relationships. ESG factors show true structural connections. Overfitting to random partitions is eliminated.

5.4 | Client Level Analysis

Client model performance helps in giving invaluable insight on the federated dynamics. Evaluation of each local model is tested on a global test set. This provides an equitable and comparable performance of all entities. The results of the five clients are

remarkably similar and high in predictive results. Client 2 has the highest accuracy in terms of local performance. Local training procedures are strong as indicated by narrow ranges of performance. The federated model obviously performs better as opposed to the average local client results. This shows the actual rewards of smart collaborative learning practices. Global model $\hat{y}_{\text{global}}$ is used to successfully combine knowledge based on weighted local predictions.

Client 4 contributes the most due to its superior validation performance. The aggregated model effectively mitigates individual client weaknesses. This analysis validates the federated framework's design for practical deployment. FedDWA scalability ensures beneficial network expansion effects. Additional clients increase global data diversity significantly. Lower-quality local updates cannot compromise the model. Sample-size-based aggregation creates undue influence risks. Large but noisy clients gain excessive weight traditionally. FedDWA employs performance-centric aggregation instead. This ensures high global R^2 maintenance. Performance remains robust in larger topologies. Heterogeneous client environments are handled effectively.

6 | XAI

The last phase of the methodology will provide transparency of the model and trust among stakeholders. Complex ensemble models are frequently opaque and black box predictors. This obscurity inhibits effective use in most of the regulated financial areas. The framework combines two explainable artificial intelligence research methods which are complementary to each other. SHAP gives global and local explanations using game theory. LIME supports very local interpretability of predictions of individual ESG scores. SHAP values aim to assign a given output of prediction to each individual input feature. The model agnostic PermutationExplainer is used in the implementation to make it more robust. LIME estimates the models that are

TABLE 6 | Performance comparison against baselines under stochastic noise and temporal scenarios.

Method	RMSE	MAE	R^2 score	MedAE	Exp. var
<i>Validation: Noisy Injection (Sigma = 0.1)</i>					
Proposed: HSE-ESG	1.9786	1.3992	0.9828	1.0402	0.9828
Proposed: Fed-ESGNet	2.2152	1.5882	0.9784	1.1700	0.9784
FedAvg	2.6296	1.8763	0.9696	1.4464	0.9696
FedSGD	2.6296	1.8763	0.9696	1.4466	0.9696
FedProx	2.6301	1.8764	0.9696	1.4445	0.9696
FedNova	2.6437	1.8864	0.9693	1.4504	0.9693
FGB	2.6449	1.8834	0.9693	1.4365	0.9693
<i>Validation: Temporal Split (Test: 30%)</i>					
Proposed: HSE-ESG	2.2828	1.5981	0.9746	1.1463	0.9750
Proposed: Fed-ESGNet	2.4974	1.7109	0.9696	1.1770	0.9701
FGB	3.1454	2.0855	0.9518	1.4131	0.9529
FedProx	3.1536	2.0865	0.9516	1.4106	0.9527
FedAvg	3.1574	2.0872	0.9515	1.4109	0.9525
FedSGD	3.1575	2.0872	0.9515	1.4109	0.9525
FedNova	3.1708	2.0936	0.9510	1.4172	0.9521

complex locally by a simple linear surrogate. SHAP and LIME offer the complete and needed model explainability.

6.1 | SHAP Analysis

SHAP analysis provides unified explanations based on cooperative game theory. The Shapley value fairly attributes a prediction to each input feature. This method satisfies desirable properties of local accuracy and consistency. For a given model f and instance x , the SHAP value ϕ_i is

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)], \quad (27)$$

$f(S)$ denotes the prediction made by the model on a subset of features S only. This formulation allows a mathematically rigorous attribution scheme. It deploys PermutationExplainer to feature estimate using a model that is agnostic to data. This explainable is useful at approximating Shapley values by perturbing input feature values. It quantifies an average marginal contribution of each single input feature. The visualization brings out the importance of features globally and directional models effects clearly. BoardDiversity measures show a positive and sustained directional effect in general. The total contributions of features would be equal to the end model prediction result.

SHAP gives single explanations that are founded on the principles of cooperative game theory. The beeswarm summary plot presents a worldwide feature influence among instances. All these effects are pointed out in Figure 4. In particular, horizontal axes show the effects of variables on the predictions of target governance. Besides, there is a positive directional effect shown by higher management scores. Local waterfall plots also break down individual predictions into individual contributions. The particular visualization causes the identification of why the model achieves the following values. These insights are used to increase model transparency by the meta learner. These tools of interpretability provide uniform estimates to different financial regulators. Rigorous attribution schemes are the measures of average marginal contributions of all inputs.

6.2 | LIME Analysis

LIME gives extremely localized explanations of the individual model predictions. This method is a local surrogate approximation of complex models that can be interpreted. LIME fits simple linear model on the basis of a particular instance of interest. Surrogate model is trained on a perturbed dataset that is close to that instance. The locality is characterized by a measure of proximity π_x of an instance x . A fidelity objective function $\xi(x)$ is minimized by the following explanation:

$$\xi(g) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g). \quad (28)$$

The complex prediction model, which should have been used in the first place, is denoted by f in this equation. The g denotes an interpretable class G model. The loss $\mathcal{L}$ is a measure of fidelity between f and g . The complexity of the explanation g is regulated by the regularization term, $\Omega(g)$. The execution provides explanations of the instances of the selected test sets. Particular attention is given to instance five in the output of analysis. The

description contains the weights of local features and one of their directional impacts. Positive weights show the characteristics that enhance the predicted ESG score. Negative weights are those features that have a negative effect on the value of governance as predicted. This gives granular transparency in understanding and validation of stakeholders. LIME completes SHAP by providing instance specific intuitive crystal.

At this particular point, LIME offers highly localized explanations of the predictions of each model. Simple linear models make an approximation of complex results on specific cases instead. All these results are represented in Figure 5. Positive weights are the ones that help to detect features that will positively respond to the predicted governance scores. On the other hand, negative weights emphasize those things that have a negative influence on the predictive governance. Such an approach provides granular visibility to various validations by the stakeholders. It also gives the weights of features in Cases 5 and 15. Green bars represent the positive impacts whereas red bars represent the reductions. This localized method supplements SHAP since it offers instance specific crystals. Complex predictions that are made through such surrogate models are fully interpretable.

6.3 | Limitations

The evaluation for this study will make use of just one dataset from Refinitiv. Future work should involve more than one alternative dataset. For our federated simulation, we assume perfect distributions for independent data. FedDWA demonstrates resilience to variance within partitioned datasets. Further research must evaluate extreme non-IID conditions thoroughly. Extreme feature skewness represents a critical test scenario. Highly imbalanced class distributions also require careful examination. These conditions may exist across institutional data silos.

Institutional data in the real world are not homogeneous and not identical. The present system is limited to processing only the numerical ESG factors. The system does not make use of the unstructured information present in sustainability reports. The model has been trained on the past data up to the point of 2022. The applicability of the model for the present conditions after the pandemic is not verified. The HSE-ESG model architecture is complex. This complexity creates a larger parameter payload overall. Shallow architectures require significantly fewer parameters instead. FedDWA optimizes the number of communication rounds effectively. Initial synchronization demands more robust bandwidth allocation. The stacking ensemble exceeds simpler linear federated models.

The integration of SHAP and LIME generates necessary granular insights. These methods provide post hoc interpretability rather than intrinsic model transparency. Regulatory bodies often require exact trace-level logic specifications. This differs fundamentally from probabilistic approximations offered by ensemble methods. Such distinction represents a clear boundary in the current scope. The framework is optimized primarily for predictive precision tasks. It serves internal risk assessment purposes effectively and efficiently. Automated regulatory reporting demands mandatory white-box traceability standards. The current system does not satisfy these stringent requirements.

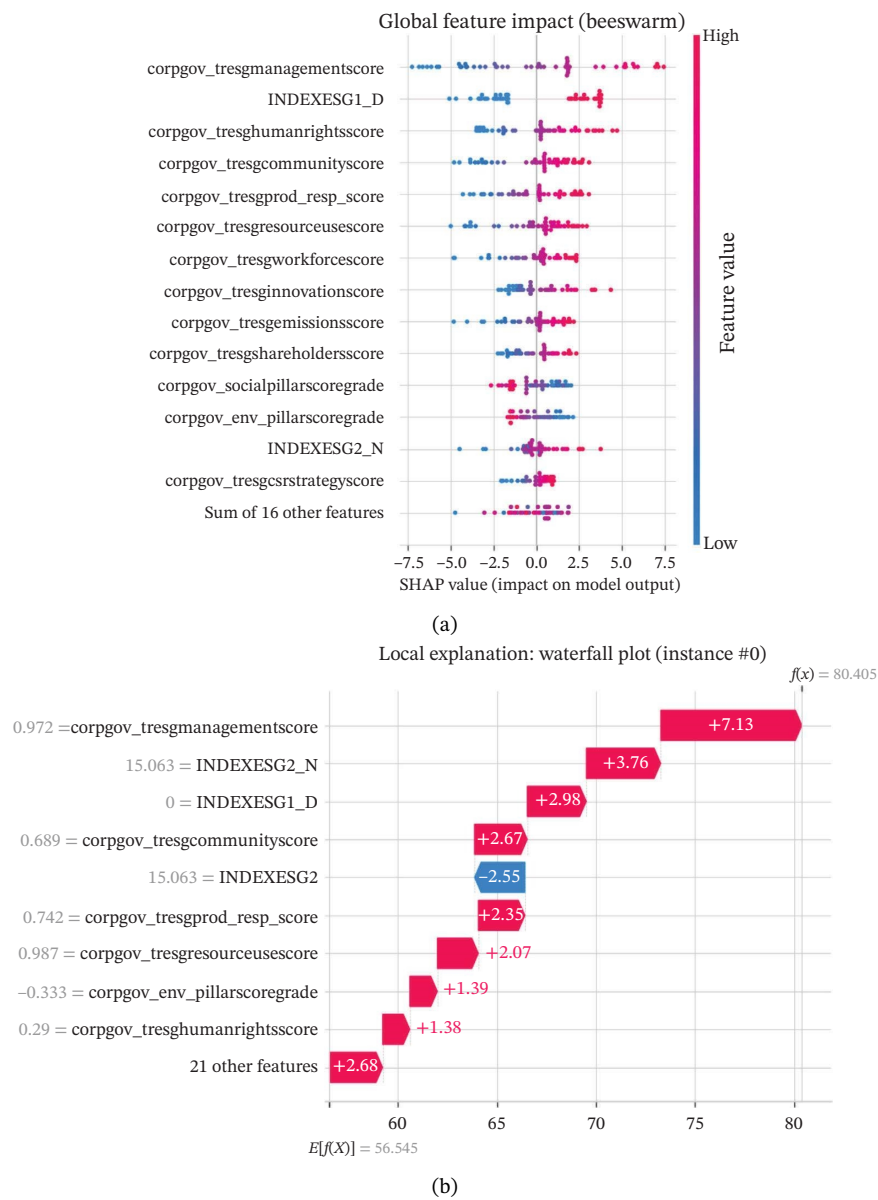


FIGURE 4 | SHAP feature importance. (a) SHAP feature impact. (b) SHAP waterfall plot.

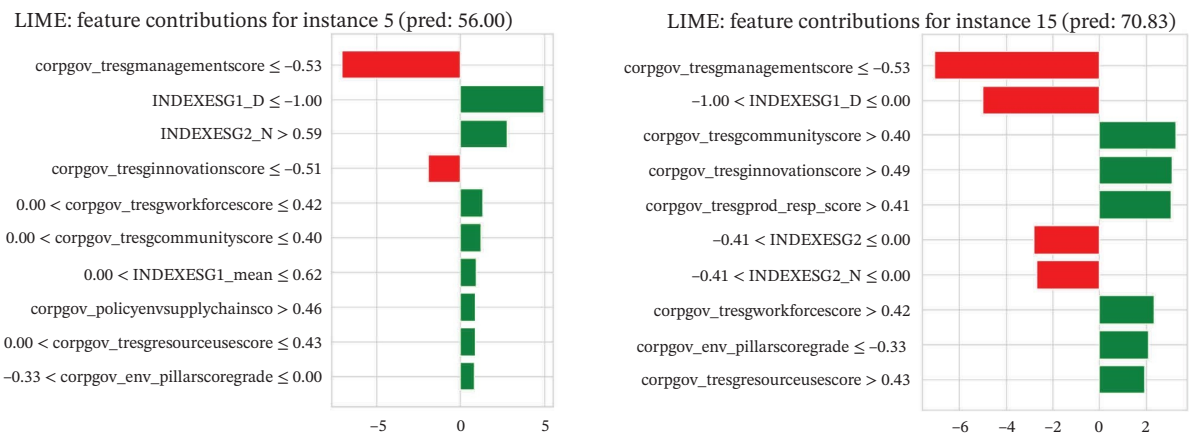


FIGURE 5 | LIME feature importance.

However, it could pose limitations in terms of deployment in environments that have constrained resources. The federated aggregation is simulated using an approach that involves five simulated clients. Larger networks could pose different scalability issues. The current simulation validates institutional-scale silos effectively. Future studies should explore massive-scale federated environments. Hundreds of small-to-medium enterprise nodes require examination. This enables deeper investigation of communication latencies. Potential weight dilution effects must be analyzed. These effects arise with larger client populations. Performance-based weights distribute across expanded networks. The preprocessing of data involves median imputation for missing data points in the data. This method can fail to model complex patterns that may exist in data points that are considered missing.

6.4 | Future Work

Future research will require the framework to be tested and validated with a variety of real-world datasets. It is necessary to test with nonindependent data distributions. The incorporation of text data sources from company reports is an important direction for research. Extracting crucial information related to ESG factors using natural language processing (NLP) could help obtain significant insights related to ESG sentiments. The application of the proposed model over live data streams could be a promising venture as well. Real-time ESG risk analysis could be achieved this way. Other sectors of finance could be explored using this proposed architecture as well.

Also, fully homomorphic encryption provides a high level of security against data that contains confidential information. Lastly, simplifying the architecture of the proposed model may enhance efficiency in computation. This would ensure ease of use in a wide range of fields.

7 | Conclusion

This study addresses the critical intersection of accurate financial forecasting and data privacy within the realm of ESG scoring. By establishing a dual-phase methodology, the research successfully demonstrates that high-precision predictive modeling need not be compromised by the constraints of data localization and privacy regulations. The proposed centralized HSE-ESG architecture, utilizing a hierarchical Bayesian Ridge metalearner to synthesize diverse base learners (LightGBM, XGBoost, and MLP), achieved a superior R^2 of 0.9854. This establishes a new benchmark for predictive accuracy by effectively capturing nonlinear dependencies and mitigating outlier distortions through robust preprocessing and feature passthrough strategies.

Furthermore, the transition to the collaborative Fed-ESGNet framework validates the feasibility of privacy-preserving machine learning in the financial sector. The implementation of the novel FedDWA algorithm marks a significant methodological advancement; by determining client contribution weights based on local validation performance rather than sample size, the framework ensures that global model aggregation is driven by predictive quality. Empirical results confirm that the federated model retains exceptional performance ($R^2 = 0.9799$), exhibiting negligible degradation compared with the centralized baseline while maintaining strict data sovereignty.

Beyond predictive metrics, the integration of XAI techniques, specifically SHAP and LIME, ensures that the complex ensemble architectures remain transparent. This interpretability is vital for regulatory compliance and stakeholder trust, providing granular insights into feature contributions and local decision boundaries. While the current study utilizes structured data from a single provider, future research will expand the framework's scope to include unstructured textual data via NLP and explore real-time risk analysis. Ultimately, this work offers a scalable, robust, and mathematically rigorous foundation for the next generation of sustainable finance systems, balancing the imperative of accurate governance assessment with the necessity of data security.

Author Contributions

Abdul Kadar Muhammad Masum: conceptualization, investigation, data curation, methodology, supervision, and writing–review and editing. Md. Abul Kalam Azad: conceptualization, investigation, methodology, supervision, project manager, and writing–review and editing. Md. Tofael Ahmed Bhuiyan: conceptualization, data curation, investigation, formal analysis, methodology, software, visualization, and writing–original draft. Md. Abdur Rahman: conceptualization, data curation, formal analysis, investigation, methodology, validation, visualization, and writing–original draft.

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Ethics Statement

No human or animal subjects were involved.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets and code that support the findings of this study are available from the corresponding author upon reasonable request.

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